

Course Guide and Syllabus

EART23001 Atmospheric Physics and Weather

About this module

EART23001 *Atmospheric Physics and Weather* develops a physical understanding of the atmosphere, from observations and vertical structure through atmospheric dynamics, moist thermodynamics, clouds, weather systems and the global energy balance. The emphasis is on using relatively simple physical principles to interpret real atmospheric behaviour.

The weekly problem sheets are an integral part of the module rather than an optional add-on. They are designed to help you move from equations and concepts to calculations, diagrams and meteorological interpretation.

Teaching and learning

The main timetabled session is on **Thursdays, 09:00–11:00, in Schuster Moseley Theatre**. A typical two-hour session will combine:

- a focused opening lecture introducing the main physical ideas;
- time to work through selected **In Class** problems, individually or with others;
- discussion and worked examples; and
- a short wrap-up that draws the physics together and connects it to observable weather and later topics.

After the session, you should normally attempt the questions labelled **You Should Try**. Questions labelled **Extra Challenge** are optional and are there for students who would like additional depth or a more demanding application.

Teaching team and contact details

Name	Email	Location
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Assessment and formative practice

Assessment in EART23001 consists of a timed Canvas coursework assessment (**20%**) and a paper-based examination (**80%**). Throughout the module you will also have several opportunities for formative practice that do not contribute to your module mark.

Component	Weight	Format and purpose
Canvas coursework	20%	A timed online assessment completed in Canvas. Once you begin an attempt, you have 3 hours to complete and submit it.
Examination	80%	A 2-hour paper-based examination completed under exam conditions during the examination period.
Practice questions	0%	Unassessed questions provided within the teaching material to help you practise calculations, concepts and scientific reasoning.
Checkpoints	0%	Short unassessed checks on your understanding as you progress through the module. These are intended to help you identify topics that would benefit from further work.
Optional formative quiz	0%	A repeatable Canvas quiz. Each attempt contains 5 questions drawn from question banks , allowing you to encounter additional questions on subsequent attempts.

Key assessment dates

Optional formative quiz available	Thursday 29 October 2026 at 11:00
Canvas coursework available	Thursday 29 October 2026 at 11:00
Canvas coursework time limit	3 hours from the time you start
Canvas coursework deadline	Friday 13 November 2026 at 12:00
Canvas coursework results released	Tuesday 1 December 2026
Examination	2-hour paper-based examination during the examination period (exact date, time and location published centrally)

Formative practice

There are three complementary ways to check and develop your understanding during the module. None of these contributes to your module mark.

- **Practice questions** are provided within the teaching material. These give you opportunities to practise calculations and apply the physical concepts introduced in the module. Try to work through the questions yourself before consulting the answers.
- **Checkpoints** provide short opportunities to check your understanding as you progress through the module. If you find a checkpoint difficult, use it to identify the relevant material to revisit before continuing.
- The **optional formative quiz** becomes available in Canvas at **11:00 on Thursday 29 October 2026** and remains available for the rest of the module. Each attempt contains **5 questions drawn from question banks**. You can repeat the quiz, with questions drawn from the banks on each attempt, to gain additional practice.

The formative activities are intended to help you identify what you understand and where you need more practice. Attempt questions using your own knowledge and reasoning first, then use the answers or feedback to identify topics to revisit in the teaching material.

Canvas coursework

The Canvas coursework is worth **20% of the module mark**.

The assessment becomes available at **11:00 on Thursday 29 October 2026**. You may choose when to begin your attempt while the assessment is available, but once you start you have a maximum of **3 hours** to complete and submit it.

The final submission deadline is:

12:00 on Friday 13 November 2026

The final deadline does not extend your individual 3-hour time limit. You should start sufficiently early to allow yourself the full three hours before the assessment closes.

Results will be released on:

Tuesday 1 December 2026

Examination

The examination is worth **80% of the module mark**.

It is a **2-hour paper-based examination** completed under exam conditions during the examination period.

The exact date, start time and examination location will be published centrally. You should check the official University examination timetable when it becomes available.

How to use the practice sheets

Each question has one of three labels:

- **IN CLASS**: selected anchor problems that may be worked through during the timetabled session;
- **YOU SHOULD TRY**: expected independent practice after the session;
- **EXTRA CHALLENGE**: optional additional depth for students who want it.

For numerical questions, show the physical principle or equation you are using, substitutions, units and a short interpretation of the result. For conceptual questions, aim to explain the physical chain of reasoning rather than simply state a conclusion. Worked answers are provided to help you check both the method and the interpretation.

A separate *Symbols, Constants, Units and Key Equations* reference is supplied. Non-standard notation is also defined locally in the questions where it is introduced.

Indicative learning outcomes

By the end of the module, you should be able to:

1. describe the basic properties of the atmosphere;
2. apply the basic laws of physics that control the behaviour of the atmosphere;
3. describe and explain the drivers for motion in the atmosphere on synoptic and global scales;
4. explain the physical behaviour of dry and moist air;
5. describe the main cloud types and cloud and rain formation processes; and
6. explain how a combination of dynamical and thermodynamical processes leads to observable weather phenomena.

Syllabus and weekly structure

Week	Date	Topic	Lecturer
1	1 Oct	Discovering the Atmosphere 1	Dr Ian Crawford
2	8 Oct	Discovering the Atmosphere 2	Dr Ian Crawford
3	15 Oct	Physics of the Atmosphere	Prof Paul Connolly
4	22 Oct	Vertical Profiles	Prof Paul Connolly
5	29 Oct	Atmospheric Dynamics	Prof Paul Connolly
6	5 Nov	Reading Week	–
7	12 Nov	Dry and Moist Air	Prof Paul Connolly
8	19 Nov	Convection and SALR	Prof Paul Connolly
9	26 Nov	Clouds and Precipitation	Dr Ian Crawford
10	3 Dec	Radiosondes	Dr Hugo Ricketts
11	10 Dec	Weather and Forecasting	Prof Paul Connolly
12	17 Dec	Global Energy	Dr Ian Crawford

A useful weekly rhythm. Attend the Thursday session, work through the selected In Class problems, complete the You Should Try questions afterwards, then use the Canvas quiz as a short check of your understanding. Return to the worked answers when you have made a serious attempt yourself.
